ONE INVESTIGATION / FIVE DEPTHS / TWO OPEN QUESTIONS

The missing recipe.

We can replay how many cryptographic curves were made.
Why were some candidates skipped?
Where did the starting points come from?

19

eligible candidates in four reconstructed runs: eight published, eleven skipped.

126

bits in the x-coordinate of K-163's subgroup half-point, on a 163-bit curve.

148

shared bits in four Certicom prime-curve half-point coordinates.

Cryptographic parameters are public ingredients, not secret keys. This project examines their history by replaying generation procedures and looking for reproducible mathematical fingerprints.

The through-line: reconstruct the recipe, compare what it produced, and distinguish a clue from an explanation. No tested rule explains all eleven skips; base-point patterns provide concrete leads, not a recovered historical algorithm.

An illustrated companion to ECDSA-Origins-Revealed-Followup. Source snapshot: `cede36edf99b41b0a252352f8ee82760936e03af`, 22 October 2026. Following the Five-Level Explanation Guide. Original vector artwork; no external fonts, scripts or image services.

01 / CHILD OR NEWCOMER / A COMIC

The case of the missing recipe

Meet Kit, a curious newcomer, and Ada, a mathematical detective. They are opening an old recipe notebook.

01

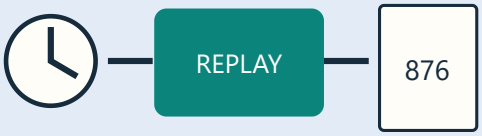
Kit: Our digital locks use public math ingredients. Who picked them?



The ingredients are called curve parameters. They are not anyone's secret password.

02

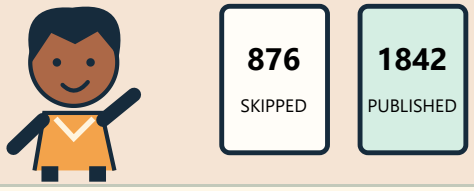
Ada: Researchers recovered much of the old recipe. Now we can replay its choices!



The same starting information produces the same sequence of possible ingredients. [S1]

03

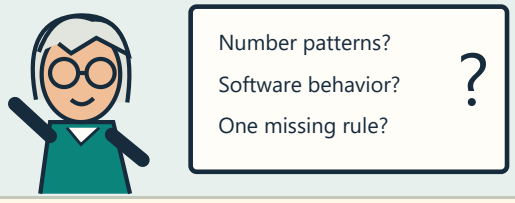
Kit: Wait! This earlier choice passes the checks we know. Why skip it?



In four runs, eight eligible choices were published and eleven other eligible choices were skipped. [S2]

04

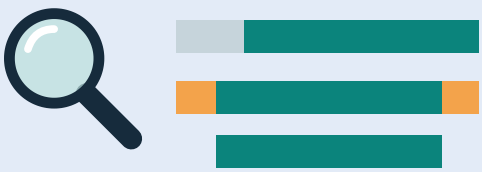
Ada: We compare their features. So far, none of our tested rules explains every skip.



Not finding the rule is useful. It is not proof that no rule exists. [S3]

05

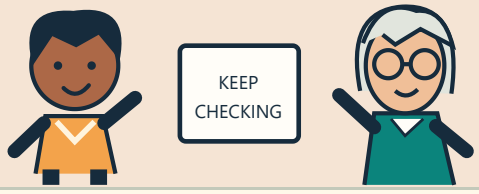
Kit: There are clues in the starting points too: a short number, and matching pieces!



Related points can reveal unusual coordinates: numbers that describe their locations. [S4, S5]

06

Ada: A clue is not the whole recipe. We need evidence, not a scary story.



This work has not shown a hidden weakness or a reason to change the curves we use.

Where the picture stops: curves are mathematical sets, not physical locks or cooking ingredients. The notebook stands for a generation program whose full history is still missing.

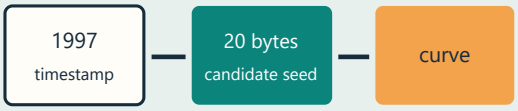
Your turn: What is the difference between finding a clue and recovering a recipe?

02 / TEENAGER OR CURIOUS BEGINNER / A COMIC

Replay. Compare. Follow the fingerprints.

01

Ada: A timestamp starts an old number generator. Its output makes candidate seeds.



A seed is an input to a public construction, not a secret signing key. Replay exposes the candidate order. [S1, S2]

02

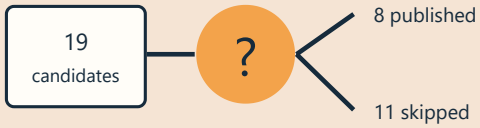
Kit: Candidate 876 qualifies. But P-256 uses 1842. Let's compare them, not guess.



Here, "qualifies" means prime group order and the reconstructed size rule, under the chosen root convention. [S2]

03

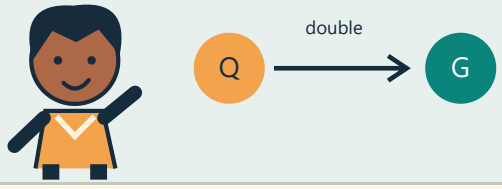
Ada: A good rule must reject the eleven skips without rejecting any of the eight published curves.



Many tests narrow the possibilities. Trying many rules also makes accidental patterns easier to find. [S3]

04

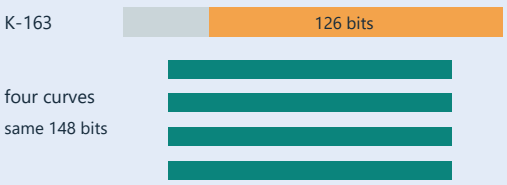
Kit: Now the base point G: a fixed public starting point. Can we work backward from it?



"Double" means add a point to itself using the curve's rules. Halving is not dividing its coordinates by two.

05

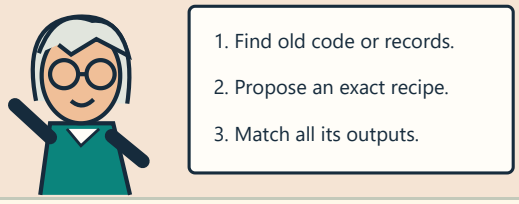
Ada: K-163 has a short half-point x. Four Certicom curves have half-point x values with a shared core.



The coordinates are reproducible evidence of structure. They do not reveal the original program by themselves. [S4, S5]

06

Kit: So: no complete explanation yet, and no demonstrated attack. What's the next test?



History needs more than a pattern match. A successful recipe must explain prefixes, tails, signs and choices too.

Why care? Dual EC showed how particular public-parameter relationships could enable a backdoor in a different construction. That motivates transparency; it is not evidence of a similar weakness here.

Your turn: Why is a rule that explains just one skipped curve not enough?

Two unknowns in one public parameter set

Prerequisites: modular arithmetic, basic probability, and the idea of an algorithm. We need a curve, a group, and a distinction between generation and selection.

A curve is a finite set with an addition rule

Over a prime field, coordinates are integers modulo a prime p. An elliptic curve can have the form $y^2 = x^3 + a\,x + b \pmod{p}$. Its points, plus an identity O, form a group. Repeated addition gives the scalar multiple $[k]P$.

A standardized parameter set includes the field, curve coefficients, a base point G, its subgroup order n, and cofactor $h = \#E / n$. In ECDSA a private scalar d gives the public key $[d]G$. Recovering the origin of G is not recovering d.

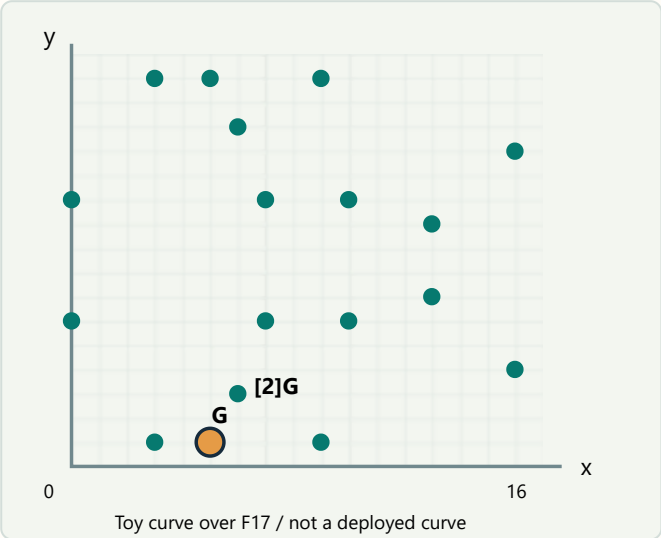


Figure 1. The toy curve has 19 points including O. Doubling G = (5,1) gives (6,3), not (10,2). Modular group operations, not ordinary geometry, govern the calculation.

Public parameters ≠ secret key

Question A: Why this curve? Question B: Why this base point?

What the historical seed does

The recovered timestamp initializes `lrand48`; for the prime-field runs, twenty top-byte outputs make each candidate seed. X9.62's SHA-1 expansion derives a value c, then $b^2 \equiv -27/c \pmod{p}$. With $a = -3$, this determines candidate coefficients when a square root exists. [S1, S2]

> **Do not confuse the two seeds:** the timestamp seeds the old pseudorandom generator; the published 160-bit seed is its output and an input to curve derivation. Neither is an ECDSA private key. A predictable parameter-generation stream is not, by itself, a signing-key attack.

Evidence: generation and viability are described in [S2]; parameter arithmetic and the corrected binary half-point are exercised in [S4].

Replay generation; test selection

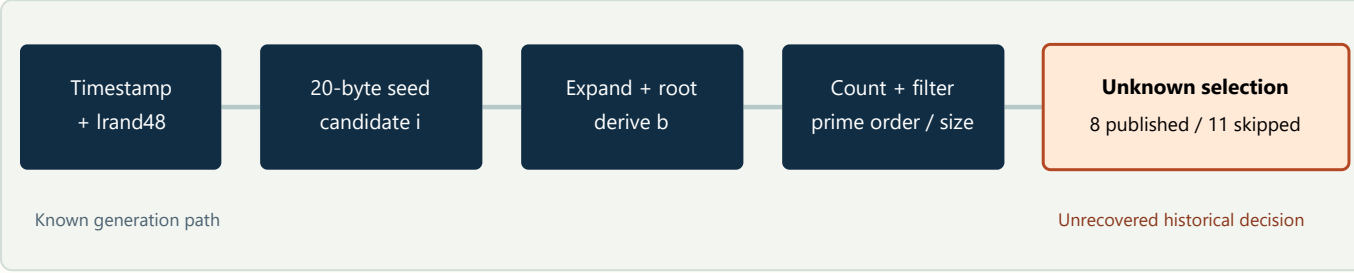


Figure 2. Generation makes a candidate; viability filters it; publication is an observed label. Reproducing the first two does not explain the third. [S2]

Eligible candidates in the four runs under the reconstructed convention [S2]

Run	Published candidate IDs	Skipped candidate IDs
P-256	1842	876
P-224	7026	777, 1485, 5736
prime192	139, 5228, 6013	2198, 5405
prime239	2185, 2320, 3048	36, 1562, 2491, 2682, 2863

Make a rule face all nineteen

For each eligible curve, compute the same features: group-order arithmetic, twist properties, CM structure, isogenies, or extension-field information. A proposed rejection rule must fit both the skipped and published labels, rather than explain just candidate 876.

The reported sweep found no such separator. CM class-number rankings even point in opposite directions in P-224 and P-256. This rejects particular explanations, not every possible hidden criterion. [S2]

Trace one actual code path

`analyze2.load` parses checked point-count logs, rejecting failures, duplicates and missing end markers. `cover.py` joins these orders with candidate seeds and coefficients, computes a Boolean predicate matrix, and searches unions of rules. Its output is the checked-in cover report. [S3, S6]

The expensive historical step is point counting and factorization. Once records exist, feature comparison is cheaper, but searching more rules increases overfitting risk.

> **Alternative to "find a pattern": measure how easy patterns are to find.** The smallest observed union uses four rules. Among 126 relabellings that preserve run counts and final published endpoints, 88 allow a cover at least as small. That is 69.8% in this conditional comparison, not a 69.8% probability that the history was innocent. [S3]

Your turn: Why must the last published endpoint stay fixed in this comparison?

Reverse the group operation, not the coordinates

If G has odd prime order n, its unique half inside that subgroup is Q = [(n+1)/2]G: doubling Q gives [n+1]G = G. This lets us inspect a point that could have preceded G in a construction. [S4]

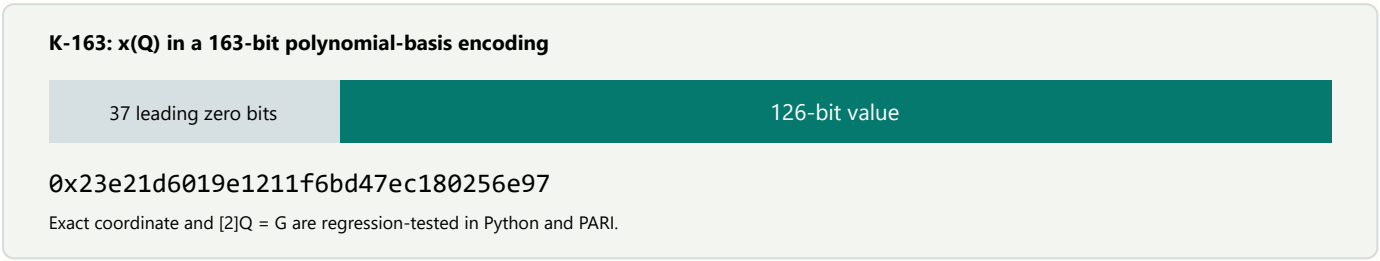


Figure 3. This is a binary-field curve, not the prime-field equation from Figure 1. Its equation is $y^2 + xy = x^3 + x^2 + 1$ over $F_{2^{163}}$. [S4]

A short coordinate is a clue

The binary scan evaluates 69,630 subgroup points per curve, 696,300 over ten curves. Under a uniform-coordinate heuristic, a specified x fitting in 126 bits has probability about 2^{-37}. Multiplying by the binary scan count gives about 5.1×10^{-6} expected x-hits. [S2]

These are structured, correlated points; other coordinates and post-hoc choices were examined. This heuristic is not a calibrated discovery probability.

Shared coordinates are a second clue

The half-point x-coordinates of secp224k1 and secp256k1 are exactly the same 166-bit integer. Four prime-field secp*k1 curves share bits 4 through 151. That strongly supports related coordinate choices, but does not recover the direction of construction, hash input, adjustment rule, or designers' intent. [S5]

A separate 92-bit prefix links the secp160r1 generator x with the secp160r2 half-point x.

secp160k1	48	ce563f89a0ed9414f5aa28ad0d96d6795f9c6	2
secp192k1	554123b78	ce563f89a0ed9414f5aa28ad0d96d6795f9c6	6
secp224k1	3b78	ce563f89a0ed9414f5aa28ad0d96d6795f9c6	3
secp256k1	3b78	ce563f89a0ed9414f5aa28ad0d96d6795f9c6	3

shared 148 bits / columns aligned, not a bit-length scale

Figure 4. Color isolates the proven common core. Prefixes and tails remain part of the historical explanation to be found; do not discard them to make a recipe appear to fit. [S5]

Halving is not the only preimage question. When a full curve has cofactor $h > 1$, inputs P with [h]P = G can differ by torsion points. The unique subgroup inverse is only one of them. All 28 rational cofactor preimages across the ten binary curves were checked in the recorded work; K-163 remains the only 128-bit-x hit. [S4]

Your turn: What would an exact recipe have to reproduce besides the green core?

Reproduce the labels before explaining them

The objective is not to classify arbitrary curves. It is to reconstruct a particular historical selection process under explicit modeling conventions.

1. **Fix the generation and root convention.** Reconstruct the timestamp-seeded stream, X9.62 expansion, and candidate order. For $p \equiv 3 \pmod 4$, all seven published coefficients match $b = (-27/c)^{(p+1)/4}$. Allowing both roots adds 23 candidates across those three runs. The nineteen-curve dataset is conditional on the chosen convention, not an exhaustive list under every implementation. [S2]

2. **Certify the group orders you rely on.** Verify n is prime, $|n - (p+1)| \le 2\sqrt{p}$, and a nonidentity P is killed by n. Since $n > 4\sqrt{p}$ in these cases, the Hasse interval contains no other multiple of n, so $\#E = n$. For composite claims, killing a point is only a consistency check. [S2, S4]

3. **Compute features without changing the labels.** Compare group-order and twist arithmetic, CM/isogeny structure and software hypotheses. Factorization-based claims need exact product verification, multiplicities, primality status and explicit unresolved cofactors; a finite ECM budget is not complete factorization. Some historical records require separate review before use as certified evidence.

4. **Calibrate a search over combinations.** Build $M_{ij} = 1$ if predicate j holds on curve i. Only predicates false on all published curves are usable in a rejection cover. Find the minimum number whose union covers every skipped curve. [S3, S6]

$$\min |J| \quad \text{subject to} \quad \forall_{j \in J}\, M_{ij} = s_i$$

$s_i = 1$ for a skipped candidate; 0 for a published candidate.

The relevant null is conditional

Within each run preserve the published count and force the final viable candidate to remain published. The number of assignments is

C(1,0) C(3,0) C(4,2) C(7,2) = 126.

Of these, 88 have a cover of size at most the observed four. This is exact enumeration for the stated label space. It is not correction for every feature, dataset convention or hypothesis selected during exploration. [S3]

A consequential data-handling boundary

The nineteen-curve comparison uses stored derived coefficients. P-224's two roots define isomorphic curves, so intrinsic CM and order properties agree; integer b residues and exact-coordinate statements need not. Keep root-dependent features explicitly labeled instead of silently substituting the standardized coefficient.

Recovered seed alignment constrains cumulative random-stream consumption at checkpoints, not the absence of every extra draw between them. [S2]

Implementation trace: `analyze2.load` → `cover.curves` → predicate matrix `M` → `min_cover` and `relabellings` → `results/cover.json`. [S3, S6]

A reproducible provenance experiment

For each standard base point, verify curve membership and [n]G = O before searching scalar transforms. Store exact coordinate encodings and the scalar witness, not just bit lengths. For binary curves this repository uses the polynomial basis specified in its parameter file. [S4]

$$Q = [h^{-1} \bmod n]G, \quad \{P : [h]P = G\} = Q + E(F_q)[h]$$

Assumes gcd(h,n) = 1. The kernel is over the rational point group, not an unspecified extension.

Test the other preimages

For K-163, h = 2 and T = (0,1) is rational two-torsion. Q and Q+T both double to G. Their x-coordinates have 126 and 162 bits. The regression tests also verify all ten binary exports and the 28 rational cofactor preimages, rather than extrapolating from the subgroup half. [S4]

The SEC prime-field report supplies a separate family of exact arithmetic witnesses. Its shared core test is $((x >> 4) \& (2^{148}-1)) = C$, with C = `0xce563f89a0ed9414f5aa28ad0d96d6795f9c6`. [S5]

Separate observation from search coverage

An earlier binary coordinate export reversed polynomial coefficients. Bit lengths survived; printed integers and search targets did not. The corrected K-163 target is regression-tested against PARI. [S4]

The SEC core search's legacy script could silently omit unavailable RIPEMD-160. Its recorded negative count is not a modern coverage manifest. Prior review fixes were ported to the separate base-point repository, not merged into this source snapshot. Do not describe those fixes as main-branch behavior. [S7, S8]

Smallest useful verification ladder

```
# From the repository root; set GP or put gp on PATH for PARI checks:
python -B -X utf8 -m unittest discover -s scripts -p test_research.py -v
# Optional: rewrites results/cover.json and notes/cover.md; may take minutes.
python -B -X utf8 scripts/cover.py
```

Do not casually rerun `secg_scan.py` expecting it to refresh the checked-in report: in this snapshot it writes a relative `secg_scan.json`. The appendix links both the inspected code and its historical outputs. [S5, S7]

> **A research extension with a falsifier:** specify a candidate string/RNG recipe and an on-curve adjustment rule before broadening the search. Require exact outputs for all four secp*k1 prefixes, cores and tails, plus the actual y/sign convention. An independently sourced historical implementation would be stronger evidence than a flexible fitted recipe.

Your turn: Which claims survive changing b to −b, and which must be recomputed?

The contribution is bounded, not vacuous

Two outputs: a conditional negative result about candidate selection, and exact structural observations about generator provenance. Neither is an attack.

Claim class	What this work supports	What it does not support
ARITHMETIC	Reconstructed viable lists; nineteen prime orders checked by primality, Hasse and annihilation. [S2, S4]	A complete reconstruction of every original filter or every possible root-handling strategy.
NEGATIVE SEARCH	No separating rule among the investigated invariant and program-model families. [S2]	Statistical equivalence of published/skipped populations, or nonexistence of an arbitrary hidden predicate.
EXACT CONDITIONAL	Four-predicate observed cover; 88/126 endpoint-conditioned assignments have cover size ≤ 4. [S3]	A posterior probability of benign selection; a confirmatory test immune to adaptive feature selection.
COORDINATE IDENTITY	K-163's 126-bit subgroup-half x; secp224k1/256k1 identical 166-bit half x; four-curve shared 148-bit core. [S4, S5]	A historical construction direction, recovered hash input, or demonstrated exploitable trapdoor.
INTERPRETATION	Related coordinate choices are strongly suggested; chosen-x/solve-y/double is consistent with the observations.	That Certicom demonstrably used that recipe, or that convenience constants establish intent.

The null's strength and its boundary

The exact conditional enumeration avoids Monte Carlo error and treats stopping endpoints explicitly. The predicate family is nonetheless exploratory and hand-selected. The minimal-union statistic is a complexity proxy for this library, not a universal description-length measure for historical software. A rule that differs by run, a variable-consumption state machine, or a point-count failure mode outside the simulated models can evade the comparison. [S2, S3]

Intrinsic versus representational evidence

Endomorphism-ring and isogeny invariants constrain the curve up to relevant equivalences. Integer coefficient factorizations, residues, and point-coordinate matches depend on the chosen model and representation. P-224's root ambiguity therefore matters to some feature families but not to the intrinsic CM comparison. Preserve both root/model provenance and exact standard-generator identities when translating an arithmetic result into a selection-history claim. The reported CM class numbers are Euler-product estimates without certified truncation bounds, not exact class-number certificates. [S2]

> **Verification status is part of the result.** This companion inspects source and recorded outputs and reruns the existing regression suite. It does not repeat historical SEA over all candidates, certify every factorization, rerun the full scalar sweeps, or independently inspect the spinout repository. Newly run checks are distinguished from historical reports in the source appendix.

What would change the explanation?

Candidate selection

An archival program, log, or independently motivated rejection rule should reproduce both the observed published choices and skipped eligible candidates, with an explicit account of root choice, timing, stopping, and RNG calls. It should predict something outside the labels on which it was fitted.

Seed checkpoint alignment alone excludes fewer state machines than "no extra calls." Seven root-convention matches constrain implementation choices without proving that the alternate root was never tried. These are constraints to carry forward, not substitutes for the missing decision process. [S2]

Base-point provenance

A common 148-bit core and a repeated 166-bit coordinate give concrete targets for historical recovery. But freedom to choose per-field prefixes, tail adjustments, y signs, encodings and counters can overfit a proposed construction. Demand an independently specified mechanism with exact output witnesses.

The all-zero-leading-bits heuristic for K-163 is a useful scale comparison, not a family-wise p-value: scalar dependence, cross-curve search, coordinate choices and post-selection all matter. The two rational halves also have different x sizes. Subgroup-only searches do not exhaust cofactor clearing. [S2, S4]

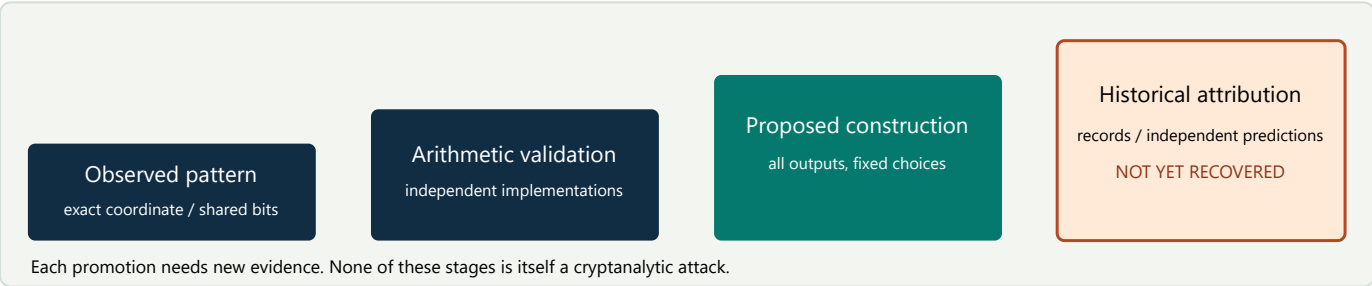


Figure 5. Distinguish exact group identities from claims about which direction the designers actually computed them in. Historical attribution is a separate evidentiary step.

Relationship to prior work and practical implications

This repository follows Anthropic's seed recovery and Valsorda's reproduction, and explicitly attributes the earlier SEC shared-generator observation to the Heninger-Scholl-Shumow Crypto 2019 rump talk. No independent novelty claim is made here: the prior literature has not been reviewed comprehensively for this companion. [S1, S2]

Nothing in these observations supplies a signing-key recovery algorithm. Dual EC's trapdoor mechanism belongs to a different construction; its relevance is the value of accountable public-parameter provenance, not a transferable accusation. There is no deployment-change recommendation from these findings.

Your turn: What independent evidence would distinguish an exact fitted recipe from the historical one?

EVIDENCE APPENDIX / SOURCE SNAPSHOT AND VERIFICATION LIMITS

What was actually inspected?

Repository sources below are pinned to `cede36edf99b41b0a252352f8ee82760936e03af`, the base of this explanation branch. The primary claim is historical provenance, not a security assessment or comprehensive code audit.

1. **[S1] Background and attribution.** Repository README and the introduction of the paper source. Those sources attribute seed recovery and six FreeLIP-derived base points to Anthropic and Valsorda. The external posts and the 2019 talk are background references, not independently reproduced historical sources in this preparation.

2. **[S2] Dataset, conventions, invariant sweep and limitations.** Paper, setup through open questions and README comparison table. Sweeps and program-model outcomes are reported results; they were not all rerun for this companion.

3. **[S3] Exact conditional cover comparison.** cover.py: min_cover, relabellings, main; recorded report. 135 nonconstant predicates, 29 usable under the true labels, minimum four, 88/126 at most four.

4. **[S4] Binary-coordinate and order verification.** test_research.py, particularly `BinaryCoordinateTests` and `EvidenceTests`; binary coordinate report. Tests exercise Python/PARI conversion, exact half coordinates, cofactor preimages and nineteen prime orders.

5. **[S5] SEC shared-coordinate observations.** SEC text report and JSON witnesses. Shared core, identical half x and 92-bit prefix are exact statements, unlike historical-construction interpretations.

6. **[S6] Input validation and code trace.** analyze2.load reads checked `data/*.orders2` records; `cover.py` joins candidate records, derives the matrix, and writes reports. This is an inspected path, not merely a directory listing.

7. **[S7] Legacy implementation boundaries.** SEC scan and core search are not the repaired review-branch versions. No new full hash or scalar search is claimed here.

8. **[S8] Scope transition.** Branch disposition records that the base-point review corrections were ported to base-point-origins, not merged here. That separate repository was not inspected for this edition.

> **New execution for this edition:** the existing 16-test regression suite was run with PARI/GP available, including cross-language and prime-order checks. The illustrated finite-field toy, displayed coordinate identities, offline links, mobile layout, and PDF export were checked separately. See the local README for commands and limits.

Editorial note: where the source snapshot infers that generators "were built" by a chosen-coordinate recipe or infers the designers' intentions, this companion uses the narrower statement justified by the arithmetic. Coordinate identities are established; the historical recipe and intent remain open.